



Pelagic Cyanobacteria

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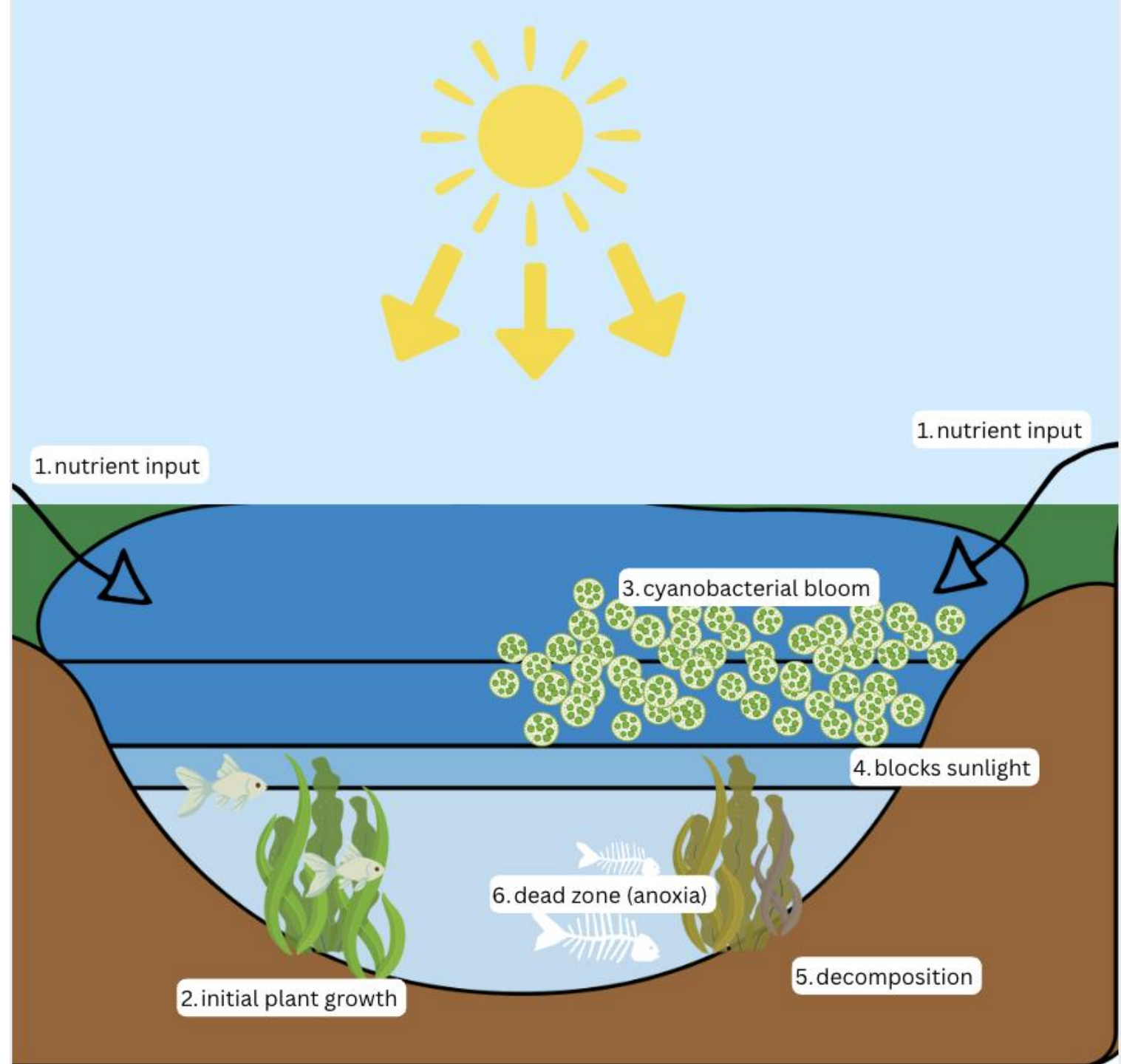


What is Pelagic Cyanobacteria?

- “Organisms in the domain bacteria able to carry out oxygenic photosynthesis with water as an electron donor and to reduce carbon dioxide as a source of carbon” (Garcia-Pichel et al., 2020)
- Pelagic vs benthic
- Source of harmful cyanobacterial blooms (cyanoHABs)

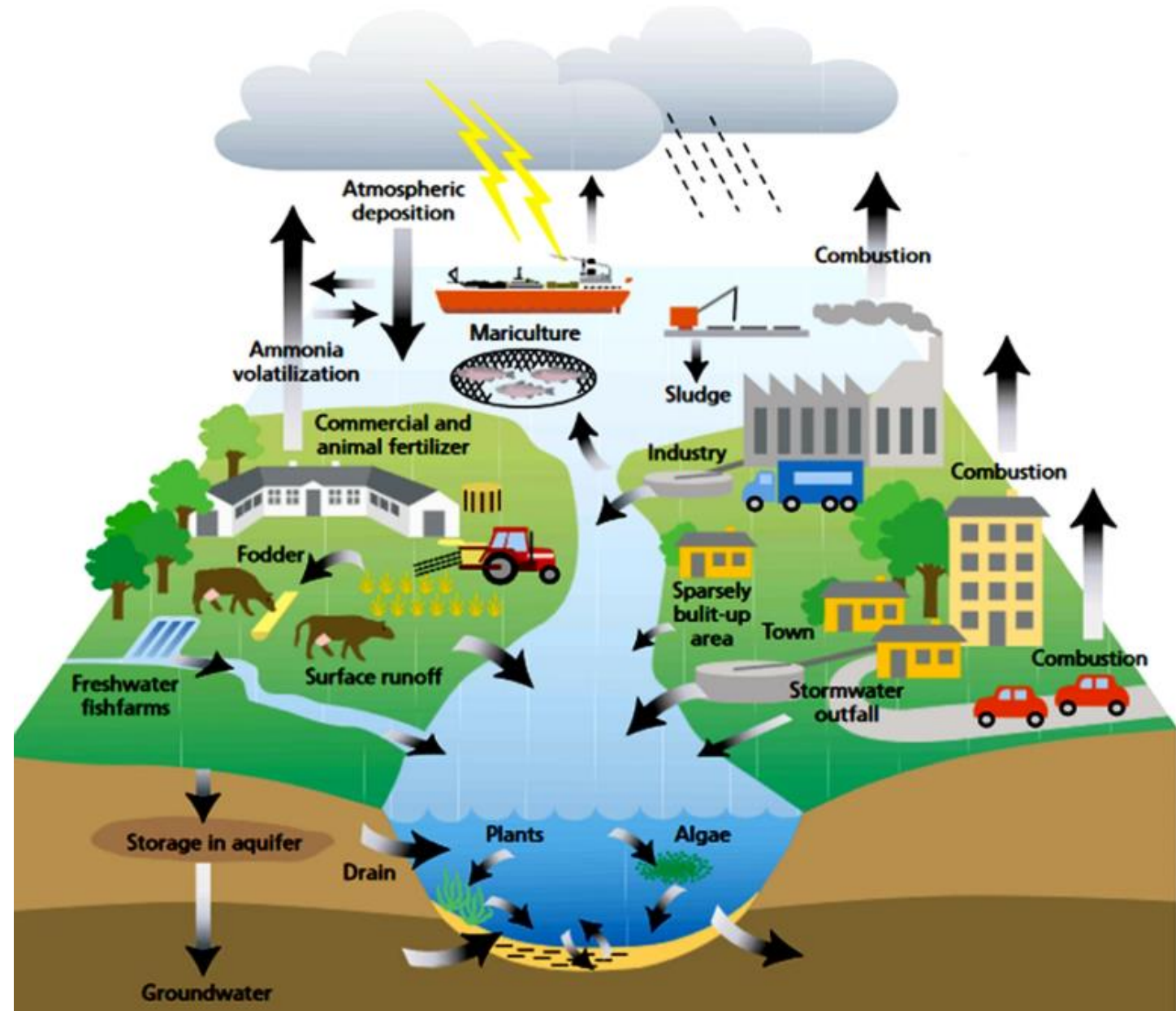
Harmful Cyanobacterial Blooms (CyanoHABs)

- Nitrogen and phosphorus increases from nutrient input (eutrophication)
- DO increases:
 - initial plant growth
- DO decreases:
 - as cyanobacteria blooms
 - when aquatic plants decompose



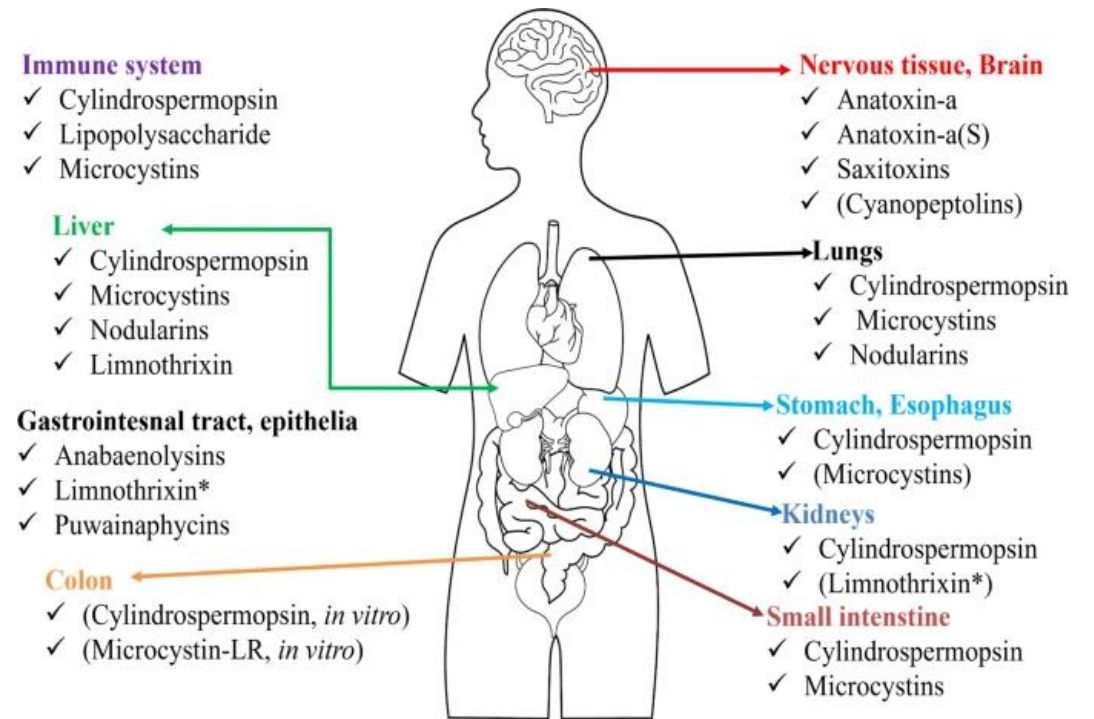
Paths to Eutrophication

- Excess of nutrients and minerals (e.g. N, P) beyond their thresholds
 - More nutrients = more growth
 - Forms harmful algal blooms
- Sources:
 - Soil erosion
 - Wastewater discharge
 - Sewage
 - Fertilizer
 - Farmland drainage



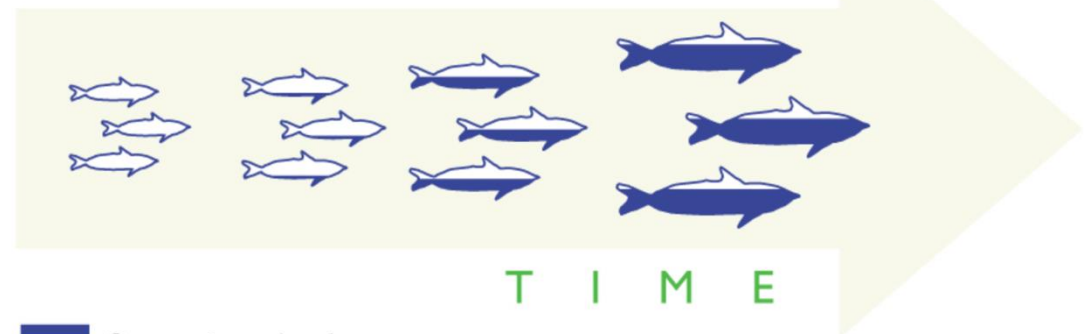
Impacts of Pelagic Cyanobacteria/HABs

- Human health:
 - CyanoHAB toxins → seafood consumption, direct exposure
 - Drinking water safety → low DO, cyanotoxins
- Ecosystem health:
 - Dissolved oxygen depletion
 - Marine organism mortality
 - Bioaccumulation of toxins



Source: Igwaran et al., 2024

Bioaccumulation

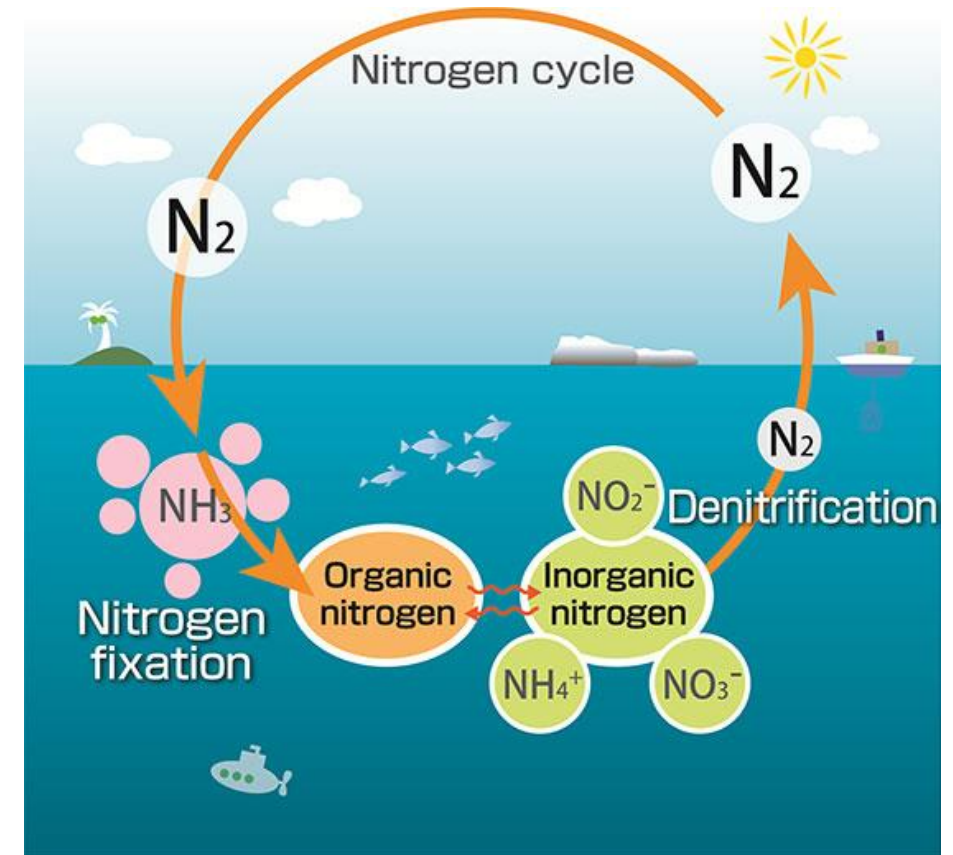


Contaminant levels

Source: van der Hoop, 2013

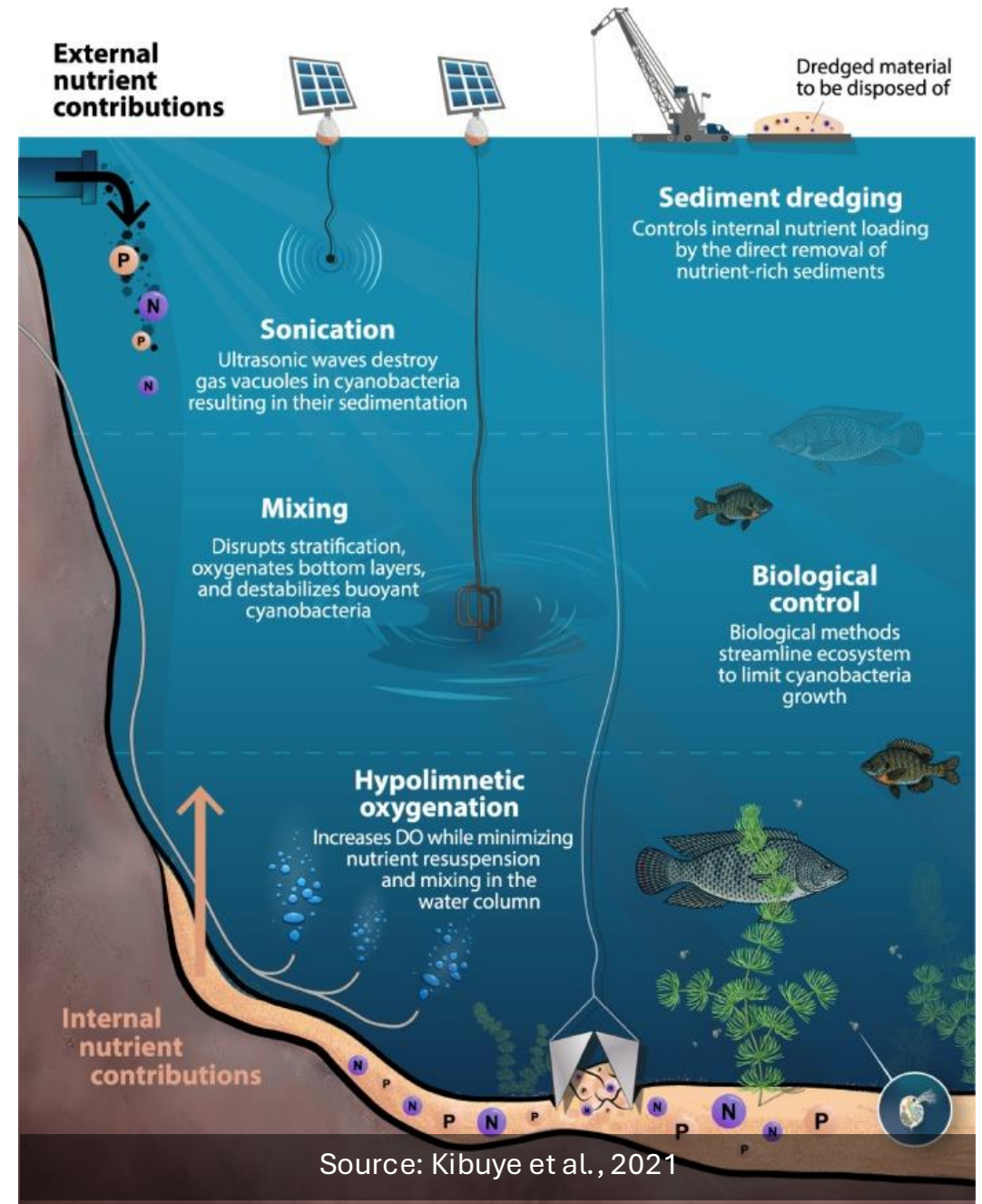
Chemistry of Pelagic Cyanobacteria

- Photosynthesis & pH
 - CO_2 uptake $\rightarrow$ O_2 production
- Nutrient chemistry (N & P)
 - NH_4^+ preferred, P storage, C:N balance
- Nitrogen fixation
 - $\text{N}_2 \rightarrow \text{NH}_3$
- Redox feedbacks
 - Bloom decay $\rightarrow$ O_2 depletion $\rightarrow$ nutrient release
- Cyanotoxins
 - Chemically active secondary metabolites



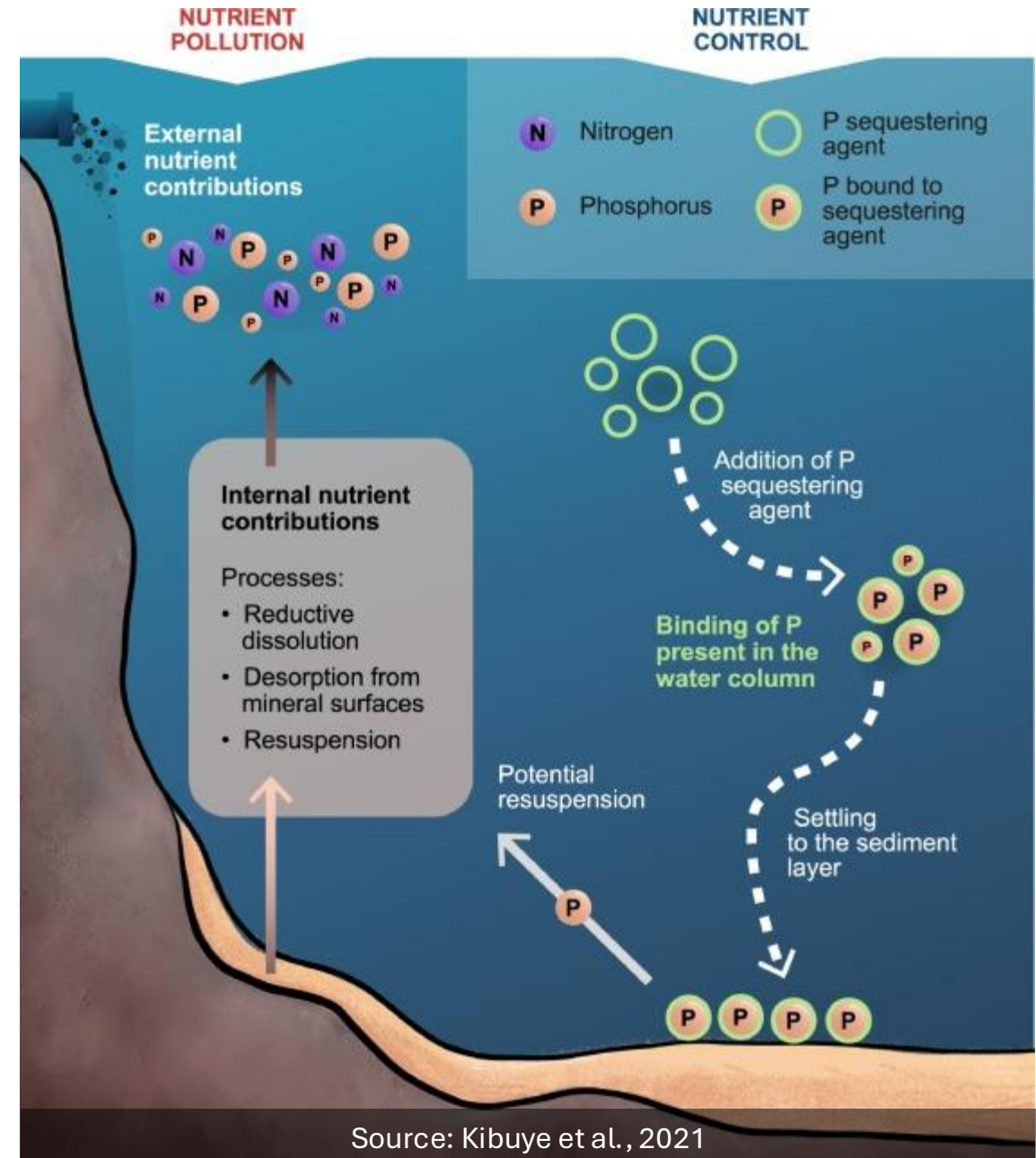
Prevention and Controls

- Physical
 - Artificial mixing
 - Hypolimnetic oxygenation
 - Sediment dredging
 - Sonication
- Biological
 - Aquatic predation
 - Aquatic plants
 - Algae consuming microorganisms



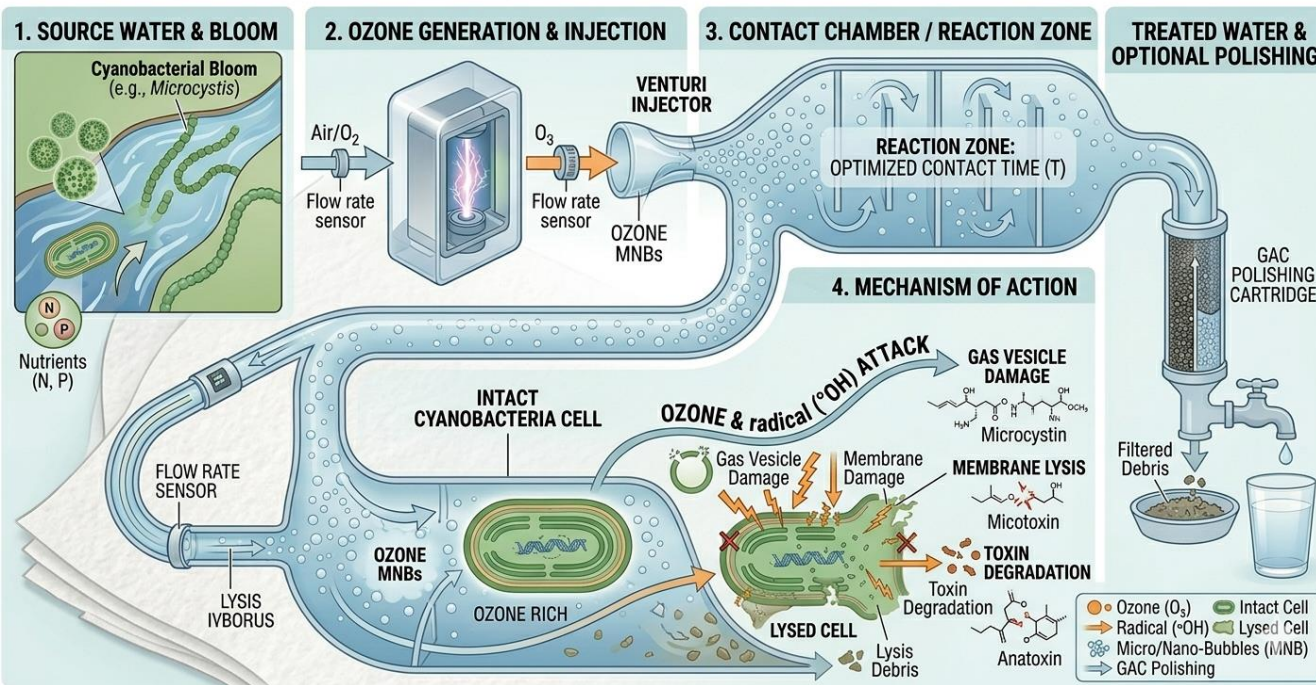
Prevention and Controls

- Chemical
 - Nutrient controls
 - Internal and external nutrient control
 - Coagulants and flocculants
 - Cyanobacteria controls
 - Algaecides
 - H_2O_2
 - Cu
 - KMnO_4

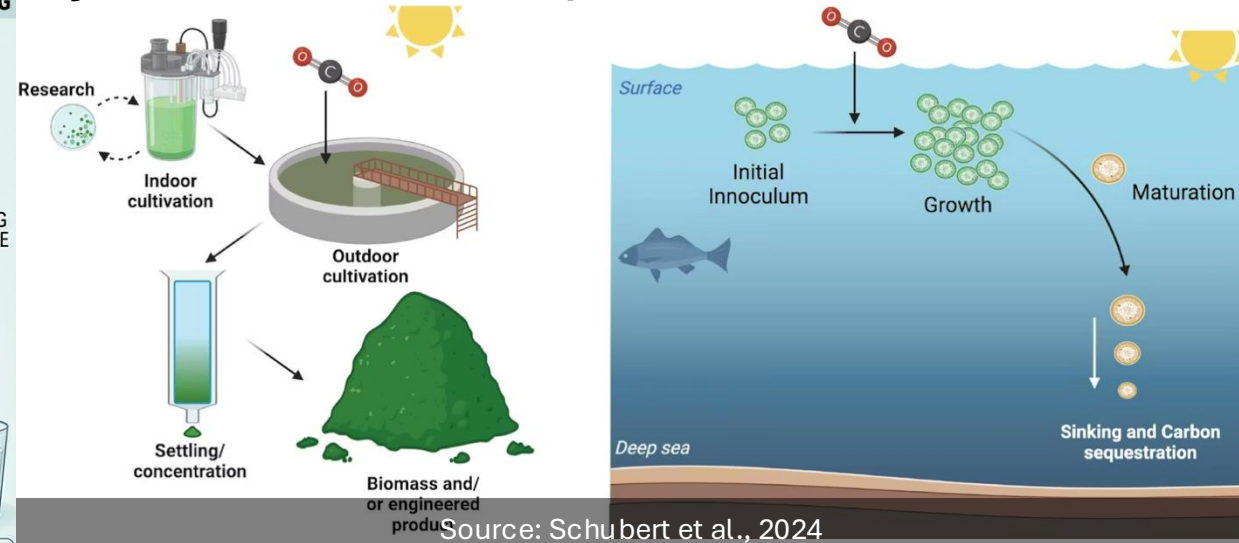


Research Developments

Ozone micro/nano bubbles

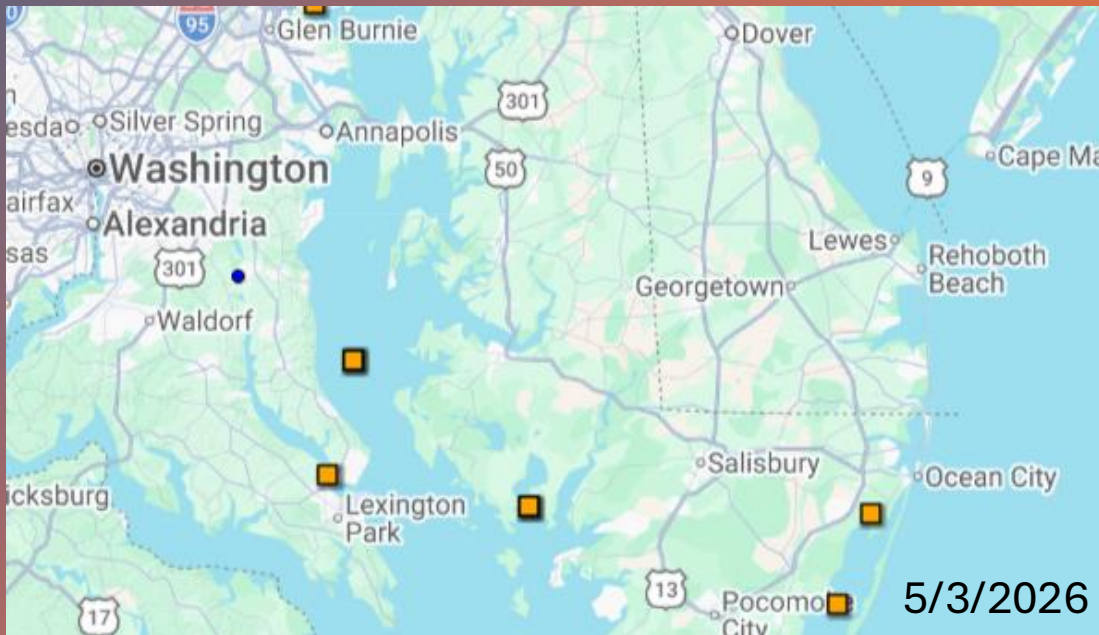
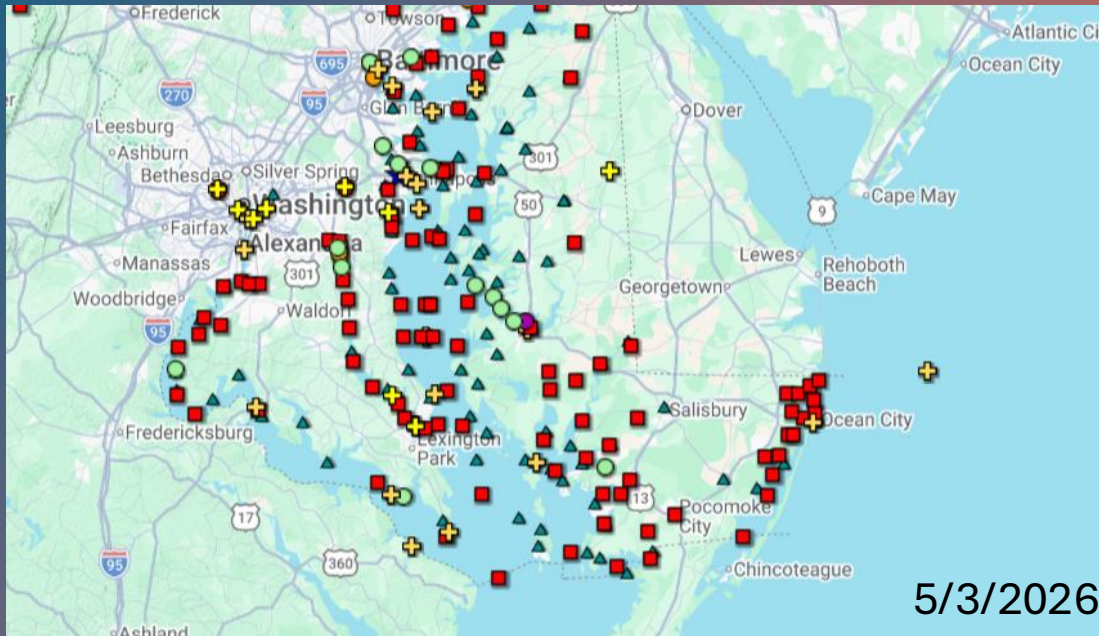


Cyanobacteria carbon sequestration



Chesapeake Bay – Recurring Blooms & Monitoring

- Nutrient pollution (N & P)
- Recurring seasonal blooms
- Seasonal hypoxia (“dead zones”)
- Ecosystem and fisheries impacts
- “Eyes on the Bay” monitoring



An aerial photograph of a coastal area featuring a large body of water with varying shades of blue and green. Several small, marshy islands and peninsulas are visible, characterized by brownish-green vegetation and intricate water channels. The text "Thank you!" and "Questions?" is overlaid in white, centered on the image.

Thank you!

Questions?

Sources

- Faith A. Kibuye, Arash Zamyadi, Eric C. Wert, A critical review on operation and performance of source water control strategies for cyanobacterial blooms: Part I-chemical control methods, *Harmful Algae*, Volume 109, 2021, 102099, ISSN 1568-9883, <https://doi.org/10.1016/j.hal.2021.102099>.
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